

Spring 1984

Problem 1 (Su83, Sp84, Fa89). Compute $\int_0^\infty \frac{\log x}{x^2 + a^2} dx$ where $a > 0$ is a constant.

Problem 2 (Sp84). For a p -group of order p^4 , assume the center of G has order p^2 . Determine the number of conjugacy classes of G .

Problem 3 (Sp84). Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous function, with $f(0) = f(1) = 0$. Assume that f'' exists on $0 < x < 1$, with $f'' + 2f' + f \geq 0$. Show that $f(x) \leq 0$ for all $0 \leq x \leq 1$.

Problem 4 (Sp84). Which number is larger, π^3 or 3^π ?

Problem 5 (Sp84). Let A and B be complex $n \times n$ matrices such that $AB = BA^2$, and assume A has no eigenvalues of absolute value 1. Prove that A and B have a common eigenvector.

Problem 6 (Sp84). Let a be a positive real number. Define a sequence (x_n) by

$$x_0 = 0 \quad x_{n+1} = a + x_n^2 \quad \text{for } n \geq 0$$

Find a necessary and sufficient condition on a in order that a finite limit $\lim_{n \rightarrow \infty} x_n$ should exist.

Problem 7 (Su80, Fa83, Sp84, Fa87, Fa96, Fa07). How many zeros does the complex polynomial

$$3z^9 + 8z^6 + z^5 + 2z^3 + 1$$

have in the annulus $1 < |z| < 2$?

Problem 8 (Sp84). Show that the system of differential equations

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

has a solution which tends to ∞ as $t \rightarrow -\infty$ and tends to the origin as $t \rightarrow +\infty$.

Problem 9 (Sp84, Sp87). Let A be a real $m \times n$ matrix with rational entries and let b be an m -tuple of rational numbers. Assume that the system of equations $Ax = b$ has a solution x in complex n -space $\mathbb{C}^n$. Show that the equation has a solution vector with rational components, or give a counterexample.

Problem 10 (Sp84). Let R be a principal ideal domain and let $\mathfrak{I}$ and $\mathfrak{J}$ be nonzero ideals in R . Show that $\mathfrak{I}\mathfrak{J} = \mathfrak{I} \cap \mathfrak{J}$ if and only if $\mathfrak{I} + \mathfrak{J} = R$.

Problem 11 (Fa81, Sp84, Sp23). If two real matrices are similar over $\mathbb{C}$, then they are similar over $\mathbb{R}$.

Problem 12 (Sp84). Consider the equation $\frac{dy}{dx} = y - \sin y$. Show that there is an $\varepsilon > 0$ such that if $|y_0| < \varepsilon$, the solution $y = f(x)$ with $f(0) = y_0$ satisfies $\lim_{x \rightarrow -\infty} f(x) = 0$.

Problem 13 (Sp84). Let I be an open interval in $\mathbb{R}$ containing zero. Assume that f' exists on a neighborhood of zero and $f''(0)$ exists. Show that

$$f(x) = f(0) + f'(0) \sin x + \frac{1}{2} f''(0) \sin^2 x + o(x^2) \quad \text{as } x \rightarrow 0$$

Problem 14 (Sp84, Sp22). Let $\mathbb{F}$ be a field and let X be a finite set. Let $R(X, \mathbb{F})$ be the ring of all functions from X to $\mathbb{F}$, endowed with the pointwise operations. What are the maximal ideals of $R(X, \mathbb{F})$?

Problem 15 (Sp84). Let F be a continuous complex valued function on the interval $[0, 1]$. Let

$$f(z) = \int_0^1 \frac{F(t)}{t - z} dt$$

for z a complex number not in $[0, 1]$.

1. Prove that f is an analytic function.
2. Express the coefficients of the Laurent series of f about ∞ in terms of F . Use the result to show that F is uniquely determined by f .

Problem 16 (Sp84). Prove, or supply a counterexample: If A is an invertible $n \times n$ complex matrix and some power of A is diagonal, then A can be diagonalized.

Problem 17 (Sp84). Prove that the Taylor coefficients at the origin of the function

$$f(z) = \frac{z}{e^z - 1}$$

are rational numbers.

Problem 18 (Sp84). Prove or disprove the following assertion: If $f : \mathbb{R} \rightarrow \mathbb{R}$ has both a left limit and a right limit at each point of $\mathbb{R}$, then the set of discontinuities of f is, at most, countable.

Problem 19 (Sp84). Let $f(x) = x \log(1 + x^{-1})$, $0 < x < \infty$.

1. Show that f is strictly monotonically increasing.
2. Compute $\lim f(x)$ as $x \rightarrow 0$ and $x \rightarrow \infty$.

Problem 20 (Sp84). Determine all finitely generated abelian groups G which have only finitely many automorphisms.